

Sentinel-2-Based Turbidity Estimation in the Yamuna River: Independent Temporal and Spatial Validation of an Etawah-Calibrated Model

Abstract

Remote sensing can complement conventional river water-quality monitoring, but models calibrated at individual monitoring locations may not transfer reliably across heterogeneous river reaches. This study developed a compact Sentinel-2-based machine-learning model for turbidity estimation using Central Water Commission observations at Etawah on the Yamuna River. Eight predictors were used: MNDWI_median, NDWI_median, MNDWI_mean, NDWI_mean, water_fraction, B11_mean, B12_mean, and B8_mean. A Gradient Boosting regression model was developed from 199 labelled Etawah observations and evaluated using grouped cross-validation by Sentinel-2 acquisition. The frozen model was then evaluated without retraining on two independent datasets: 14 unseen 2024 observations at Etawah and 812 observations from eight additional Yamuna stations represented by 521 unique Sentinel-2 scenes. The temporal test produced MAE 26.269 NTU, RMSE 35.086 NTU, R^2 0.064, and Pearson r 0.762. Spatial transfer was substantially weaker, producing MAE 68.647 NTU, RMSE 157.876 NTU, R^2 -0.107, and Pearson r 0.236. For observations ≥ 100 NTU, spatial MAE increased to 288.248 NTU, and all such observations were underpredicted. Cloud contamination increased error but did not fully explain the spatial failure because high-turbidity observations remained poorly predicted under relatively clear conditions. The results demonstrate that the Etawah-calibrated model has some temporal persistence but limited spatial generalizability across the Yamuna, particularly under extreme turbidity. Future work should emphasize multi-reach training, broader representation of extreme events, physically informed predictors, and continued independent spatial validation.

Keywords

Yamuna River; Sentinel-2; turbidity; water quality; remote sensing; machine learning; Gradient Boosting; spatial transfer; temporal validation; Central Water Commission

1. Introduction

River turbidity is an important indicator of suspended material and changing aquatic conditions. Conventional monitoring provides direct measurements but is constrained by sampling frequency and spatial coverage. Satellite remote sensing provides a complementary means of extending water-quality information through repeated spatial observations.

A major challenge is model transferability. Spectral relationships learned at one monitoring location may change along a river because of differences in sediment characteristics, channel morphology, water depth, surrounding land cover, atmospheric conditions, and the range of observed turbidity. Consequently, evaluation using only random observation-level splits can provide an overly optimistic assessment of practical generalization.

This study addresses that issue by developing a compact Sentinel-2 turbidity model at Etawah and evaluating the frozen model under two independent conditions: unseen temporal observations at Etawah and

geographically distinct Yamuna monitoring stations. The analysis also examines turbidity regime, cloud contamination, temporal matching, radiometric behaviour, and feature-distribution shift.

2. Materials and Methods

2.1 Study Area and Data Sources

Central Water Commission surface-water-quality observations were paired with Sentinel-2 Surface Reflectance Harmonized imagery (Drusch et al., 2012). The development dataset was based on the Etawah station on the Yamuna River. Independent spatial testing used eight additional Yamuna stations: AGRA(POIYAGHAT), Baghpat, Kailas Mandir, Agra, Kalanaur, Mathura (Gokul Barrage), Mawi, Vrindavan-Yamuna Expressway Link Road Bridge U/S of Mathura, and Yamuna Highway Road Bridge.

2.2 Overall Study Workflow

The overall workflow of the study is shown in Figure 1. Central Water Commission (CWC) turbidity observations were paired with Sentinel-2 Surface Reflectance Harmonized imagery through temporal matching, and predictor features were extracted from image patches around the CWC station coordinates. A Gradient Boosting regression model was developed from 199 labelled Etawah observations and evaluated using grouped cross-validation by Sentinel-2 acquisition. The frozen model was then evaluated without retraining on two independent datasets: unseen 2024 observations at Etawah (temporal validation) and observations from eight additional Yamuna stations (spatial validation).

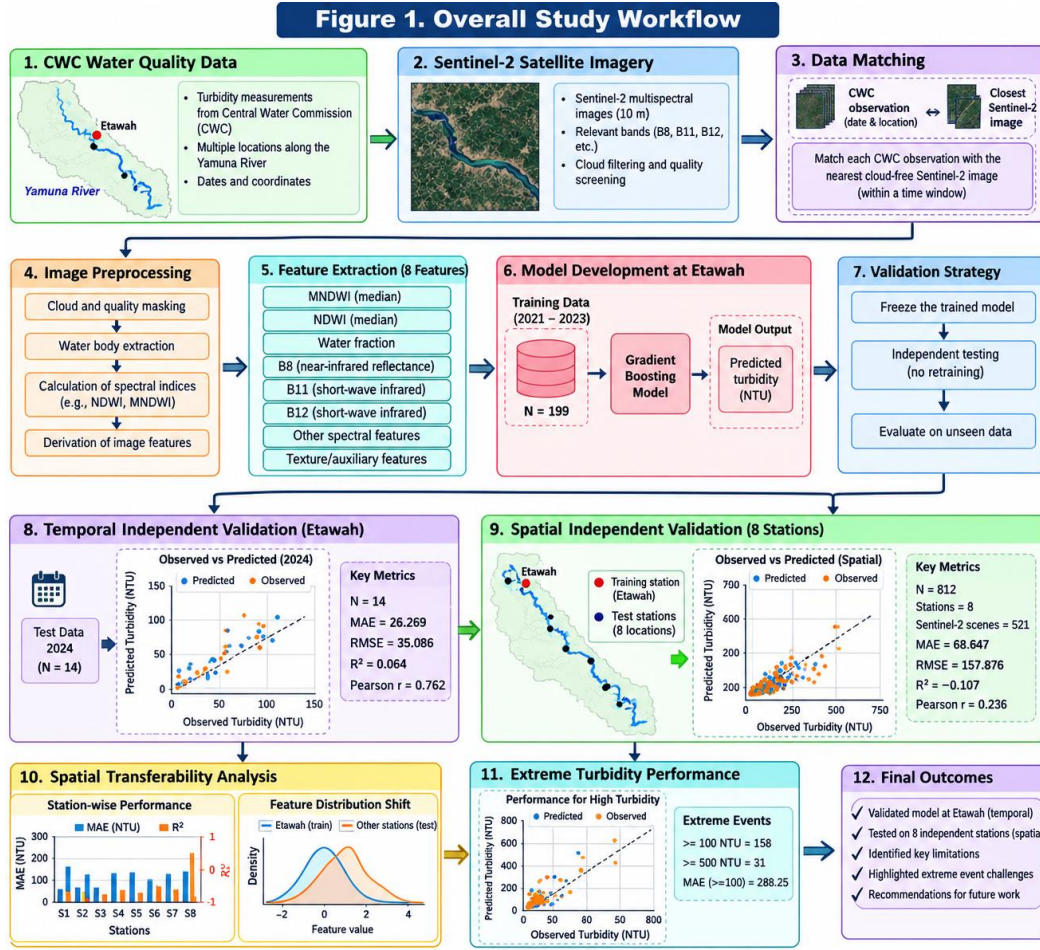


Figure 1. Overall workflow of the study, showing CWC water-quality observations, Sentinel-2 satellite imagery, temporal matching, image processing, feature extraction, model development at Etawah, and independent temporal and spatial validation.

2.3 CWC–Sentinel-2 Data Matching

Each CWC turbidity observation, identified by its station, date and time, was matched to the Sentinel-2 acquisition temporally closest to the measurement (Figure 2). Candidate acquisitions were considered within a ± 2 -day matching window around the CWC observation date and time, and the nearest valid image was selected. Valid pixels were checked, and a 256-m patch around the CWC station coordinates was then used for feature extraction. The Google Earth Engine scripts used for temporal matching and Sentinel-2 feature extraction are provided in the project repository.

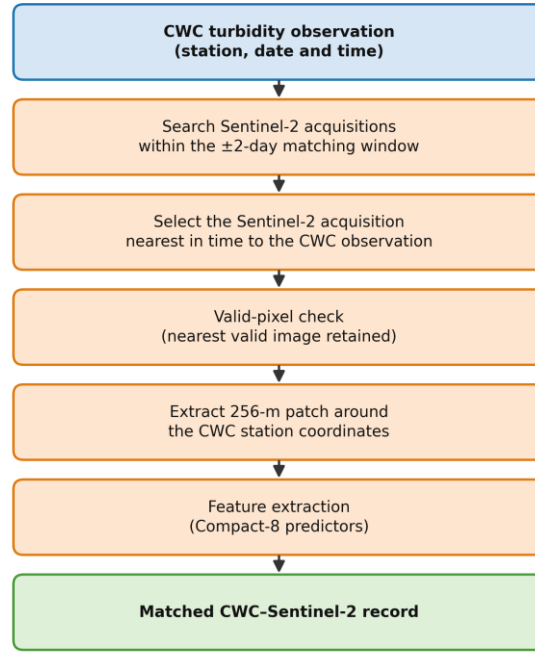


Figure 2. CWC–Sentinel-2 matching procedure used to identify the Sentinel-2 observation temporally closest to each CWC turbidity measurement.

2.4 Satellite Feature Extraction

The final Compact-8 predictor set comprised median and mean MNDWI and NDWI, water fraction, and mean B8, B11, and B12 reflectance. NDWI was calculated following McFeeters (1996), while MNDWI followed Xu (2006). Features were extracted from 256-m image patches around the CWC station coordinates.

2.5 Machine-Learning Model Development

The final estimator was GradientBoostingRegressor with `n_estimators=300`, `learning_rate=0.03`, `max_depth=2`, Huber loss, and `random_state=42` (Friedman, 2001). The development dataset contained 199 labelled observations. Five-fold GroupKFold validation was performed using Sentinel-2 acquisition as the grouping variable (Pedregosa et al., 2011).

2.6 Independent Validation Design

Temporal transfer was evaluated by training only on 2021–2023 observations and testing on unseen 2024 observations. Spatial transfer was evaluated using eight additional Yamuna stations. The final labelled spatial test contained 812 observations representing 521 unique Sentinel-2 scenes. The frozen model was not retrained or tuned during either test.

3. Results

3.1 Model Development at Etawah

The Etawah model had development (grouped cross-validation) performance of MAE 22.954 NTU, RMSE 41.599 NTU, and R^2 0.236. This performance was obtained for $N = 199$ labelled observations, and the development-model relationship is shown in Figure 3.

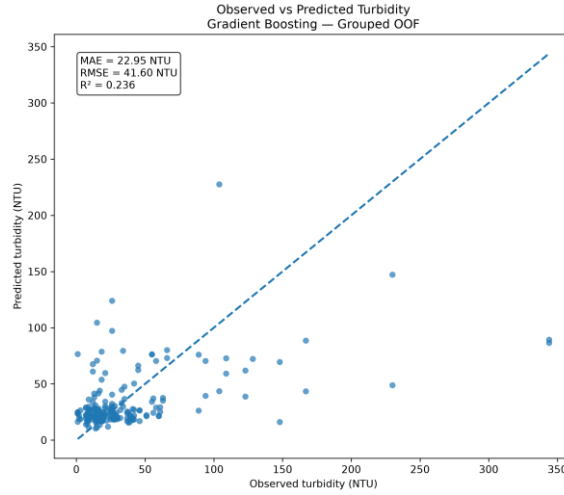


Figure 3. Observed versus predicted turbidity for the Etawah model-development dataset.

3.2 Independent Temporal Validation

The frozen model retained positive association with unseen 2024 observations (Pearson $r=0.762$), but absolute calibration was limited. MAE was 26.269 NTU, RMSE 35.086 NTU, and R^2 0.064. The negative observed-minus-predicted bias indicates average overprediction. The test sample contained only 14 observations, so subgroup results should be treated as sensitivity evidence. The independent temporal validation is shown in Figure 4.

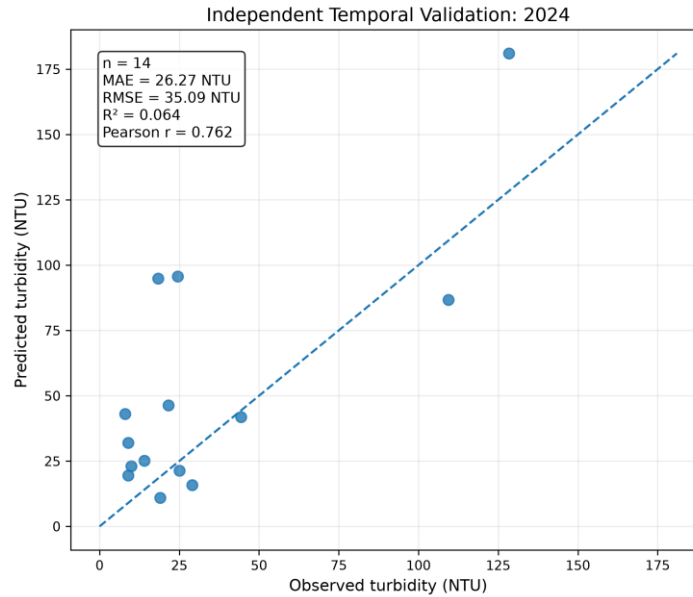


Figure 4. Independent temporal validation of the frozen Etawah model using previously unseen 2024 observations.

3.3 Independent Spatial Validation

Across 812 labelled observations from eight additional Yamuna stations, MAE was 68.647 NTU, RMSE 157.876 NTU, R^2 -0.107 , and Pearson r 0.236. The observed-minus-predicted bias was 54.999 NTU, indicating systematic underprediction across the broader spatial domain. The negative R^2 indicates that, in this spatial test, the predictions explained the observed variation worse than a simple mean-based baseline, and Pearson r = 0.236 indicates only weak linear association. The spatial validation results are shown in Figure 5, and the development, temporal and spatial results are compared in Table 1.

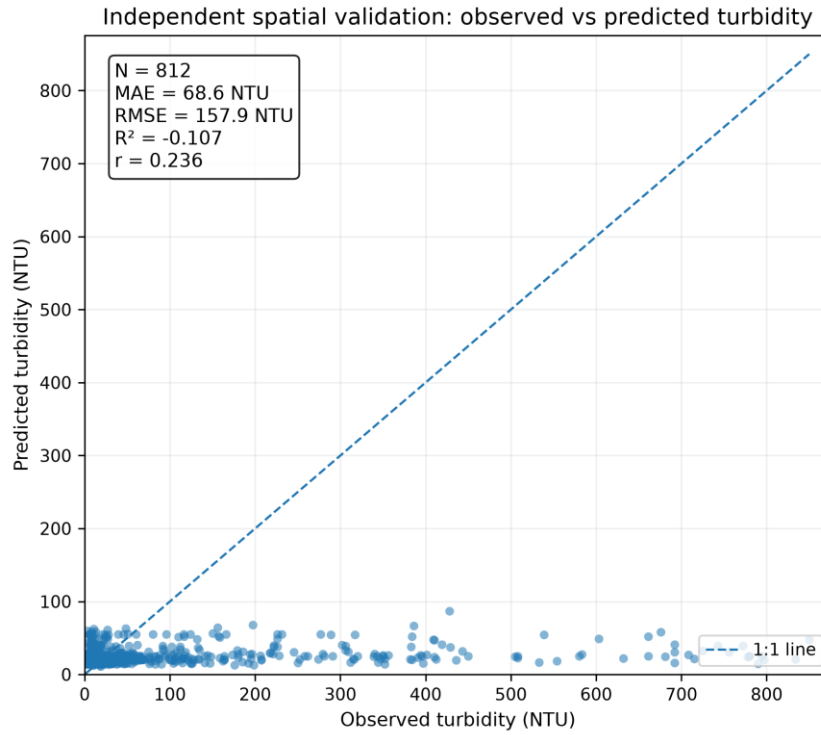


Figure 5. Observed versus predicted turbidity during independent spatial validation across eight additional Yamuna monitoring stations.

Table 1. Development, independent temporal and independent spatial validation performance of the Etawah-calibrated model (MAE, RMSE and bias in NTU; bias is observed minus predicted).

Validation	N	MAE	RMSE	R ²	Pearson r	Bias
Development	199	22.954	41.599	0.236	—	—
Temporal 2024	14	26.269	35.086	0.064	0.762	−19.140
Spatial 8 stations	812	68.647	157.876	−0.107	0.236	54.999

3.4 Station-Level Spatial Transferability

Performance for each of the eight independent stations is given in Table 2.

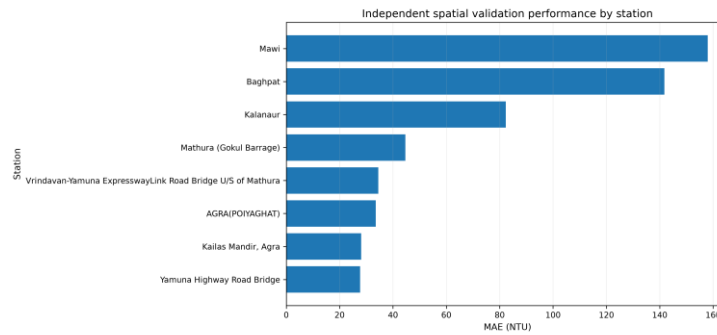
Table 2. Spatial-test performance by station (MAE, RMSE and bias in NTU).

Station	N	MAE	RMSE	R ²	Bias	r
Yamuna Highway Road Bridge	87	27.661	65.955	−0.012	4.918	0.018
Kailas Mandir, Agra	99	28.081	64.562	−0.022	15.140	0.190
AGRA(POIYAGHAT)	128	33.537	58.878	0.143	7.582	0.469
Vrindavan-Yamuna	64	34.500	92.246	0.025	18.969	0.338

Station	N	MAE	RMSE	R ²	Bias	r
ExpresswayLink Road Bridge U/S of Mathura						
Mathura (Gokul Barrage)	135	44.616	103.747	−0.069	33.178	0.262
Kalanaur	101	82.279	178.769	−0.178	73.610	0.221
Baghpat	96	141.767	258.401	−0.380	138.714	0.282
Mawi	102	157.956	268.776	−0.412	150.169	0.327

Performance differed strongly among the eight stations, and station-level transferability is summarized in Figure 6. MAE ranged from 27.661 NTU at Yamuna Highway Road Bridge to 157.956 NTU at Mawi, and R² ranged from −0.412 at Mawi to 0.143 at AGRA(POIYAGHAT). R² was positive at only two stations (AGRA(POIYAGHAT), 0.143; Vrindavan-Yamuna ExpresswayLink Road Bridge U/S of Mathura, 0.025), and the largest errors and biases occurred at Mawi, Baghpat and Kalanaur.

(a)



(b)

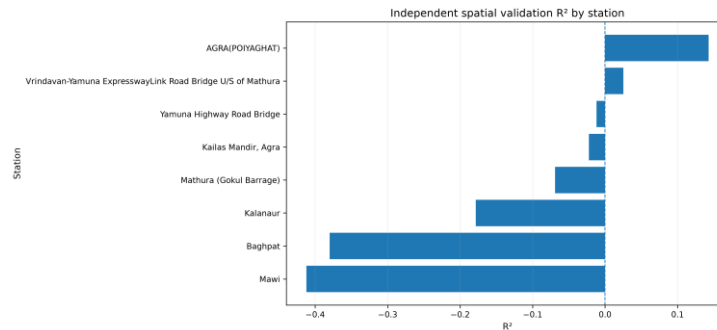


Figure 6. Station-level spatial transferability of the frozen Etawah-calibrated model. (a) Mean absolute error (MAE) and (b) coefficient of determination (R²) across the eight independent Yamuna stations.

3.5 Cloud and Temporal-Gap Diagnostics

Spatial MAE was 38.344 NTU for cloud cover ≤20% and 108.031 NTU for cloud cover >60%. Cloud therefore contributed to error, but high-turbidity observations remained poorly predicted under relatively clear conditions. Temporal-gap sensitivity similarly showed that restricting the match to ≤0.5 day did not eliminate the spatial error (MAE 65.900 NTU).

3.6 Extreme Turbidity Performance

Spatial-test performance by observed-turbidity regime is given in Table 3.

Table 3. Spatial-test performance by observed-turbidity regime (MAE, RMSE and bias in NTU).

Regime	N	MAE	RMSE	Bias	Underprediction
<10 NTU	200	17.347	19.736	−17.347	0%

Regime	N	MAE	RMSE	Bias	Underprediction
10–<20 NTU	209	8.565	11.538	–8.121	10.048%
20–<50 NTU	182	12.176	14.728	8.626	85.714%
50–<100 NTU	63	43.222	46.272	43.052	98.413%
≥100 NTU	158	288.248	355.417	288.248	100%

The proportion of underpredicted observations increased from 0% in the <10 NTU regime to 98.413% in the 50–<100 NTU regime and 100% in the ≥100 NTU regime.

The spatial dataset contained 158 observations ≥100 NTU and 31 observations ≥500 NTU. All observations ≥100 NTU were underpredicted. For observations ≥100 NTU, MAE was 288.248 NTU and RMSE was 355.417 NTU. The frozen Etawah-calibrated model therefore had substantial difficulty transferring to very high turbidity conditions. Extreme turbidity performance is shown in Figure 7.

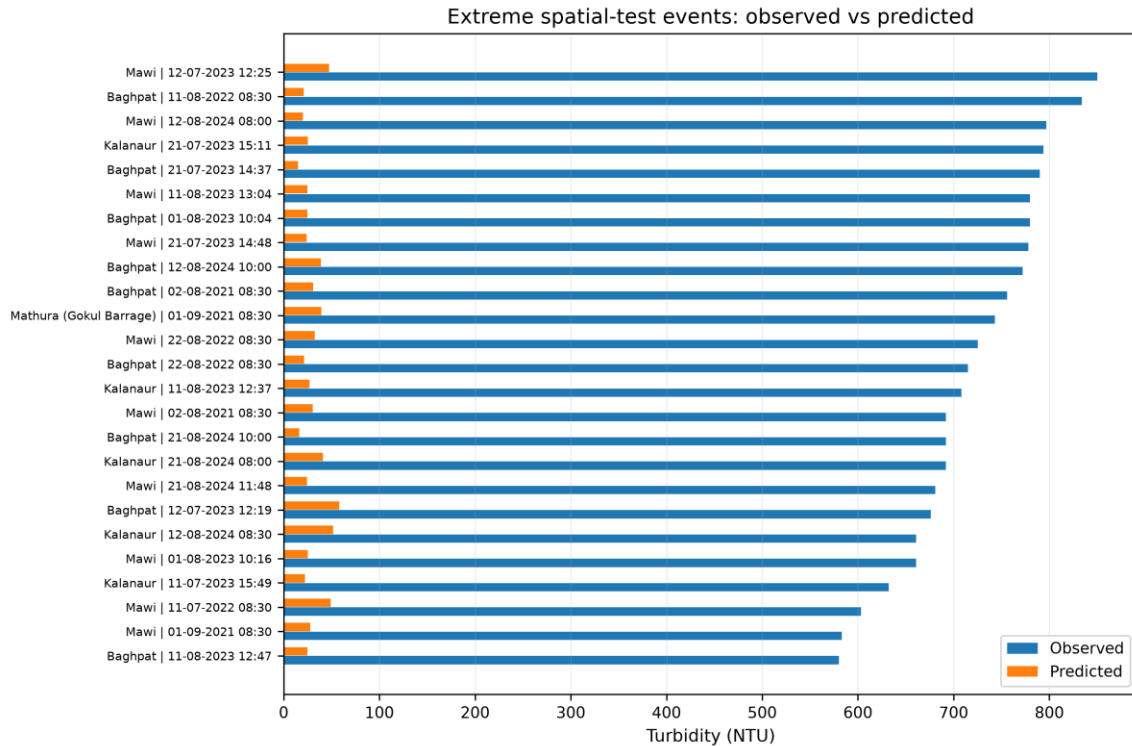


Figure 7. Model performance during extreme turbidity observations (≥100 NTU) in the independent spatial test.

4. Discussion

4.1 Temporal Versus Spatial Transfer

The central finding is the contrast between temporal and spatial transfer. The frozen model retained Pearson $r=0.762$ on unseen 2024 Etawah observations but only $r=0.236$ across additional Yamuna stations. This indicates some temporal persistence at the development site but limited geographic invariance.

The negative spatial R^2 should not be interpreted as evidence that Sentinel-2 cannot support turbidity estimation. Rather, it demonstrates that a compact single-reach calibration is insufficient for heterogeneous river-wide transfer. A future multi-reach model should span the full turbidity range and retain genuinely spatially independent validation.

4.2 Extreme Turbidity and Model Underprediction

Extreme turbidity was the dominant spatial failure regime. Predictions were compressed toward the lower portion of the observed range, producing severe underprediction at 100–850 NTU. The independent spatial dataset exposed turbidity regimes that were insufficiently represented in the Etawah development data.

4.3 Effect of Cloud Contamination

Cloud contamination was a contributing factor but not the dominant explanation. Several extreme failures occurred under low or moderate cloud cover, so stricter cloud screening alone would not solve the transfer problem.

4.4 Feature Distribution Shift and Spatial Transferability

The independent spatial dataset showed moderate covariate shift relative to the Etawah training data, including higher B8, B11, and B12 means and downward shifts in NDWI/MNDWI (Figure 8).



Figure 8. Distributional differences between the Etawah training data and the independent spatial test data for the satellite-derived model features.

These shifts are consistent with heterogeneous river and environmental conditions, but do not by themselves explain the extreme underprediction.

5. Limitations

The temporal test contains only 14 observations. The spatial test contains heterogeneous turbidity regimes, including 31 observations above 500 NTU. CWC and satellite observations are not perfectly simultaneous even after verified nearest-image matching. Cloud contamination and atmospheric effects remain relevant. The Compact-8 feature set also excludes flow, sediment composition, channel morphology, water depth, meteorological conditions, and other potentially informative environmental variables.

The current model should not be used as a Yamuna-wide operational turbidity estimator without additional multi-station training and independent validation.

6. Future Work

Future research should expand the CWC observation record available for model development and evaluation, including more Yamuna stations, more seasons, and a wider range of hydrological conditions. Additional high-turbidity observations are particularly needed, because extreme turbidity was the dominant failure regime in the independent spatial test.

Multi-station and multi-reach model development should be explored, together with additional machine-learning approaches and additional physically meaningful satellite predictors. Improved cloud and image-quality screening and seasonal analysis could also be investigated. Any such models should continue to be assessed with independent temporal and spatial validation. These directions are proposed for future study and have not been completed in the present work.

7. Conclusion

A compact Sentinel-2 Gradient Boosting model calibrated at Etawah showed moderate temporal association with unseen 2024 observations but poor spatial transfer across eight additional Yamuna stations. The spatial test produced MAE 68.647 NTU, RMSE 157.876 NTU, $R^2 = -0.107$, and Pearson $r = 0.236$. Extreme turbidity was the dominant failure regime, with MAE 288.248 NTU for observations ≥ 100 NTU and 100% underprediction in that group.

The study demonstrates the importance of genuine spatial and temporal independence in remote-sensing water-quality validation. Future work should use multi-station training, broader extreme-event coverage, additional physically meaningful predictors, and continued independent spatial validation.

8. Data and Code Availability

The analysis scripts, Google Earth Engine processing scripts, model configuration, derived results, and publication figures are available through the project's public GitHub repository: [GITHUB REPOSITORY LINK](#). The repository includes the scripts used for CWC–Sentinel-2 temporal matching, Compact-8 feature extraction, model development, independent temporal validation, independent spatial validation, diagnostic analyses, and figure generation. Original Central Water Commission observations are subject to applicable data-access and redistribution conditions and are therefore not redistributed as raw data.

9. References

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